

FMCG Demand Forecasting & Promotion Effectiveness

Walmart weekly sales — 421,570 store-department records, 45 stores, 81 departments, 2010-02-05 to 2012-10-26

45 / 81	+7.1%	+91%	4-week moving average
Stores / Depts	Holiday sales uplift	High vs low markdown (holiday)	Best forecast baseline

So what: Markdown intensity correlates with much higher sales, but markdown timing itself clusters around naturally high-traffic weeks — this overstates true causal lift. Don't scale promo budget chain-wide off this correlation alone; run a matched-store test before committing incremental spend.

Key Findings

- Holiday weeks lift sales only +7.1% on average — smaller than intuition suggests, and concentrated in Thanksgiving/Christmas rather than spread across all flagged holidays.
- High-markdown weeks show +76% higher sales (non-holiday) and +91% (holiday) vs low-markdown weeks — correlational, confounded by promo timing.
- Store Type A stores average \$20.1K/week vs Type C's \$9.5K/week (2.1x gap) — but the gap tracks store size almost linearly; Type format itself adds little independent lift.
- Forecast baseline: 4-week moving average (WMAE 55,644) narrowly beats seasonal-naive (WMAE 56,385) on a 12-week holdout, evaluated with the competition's holiday-weighted metric.

Charts

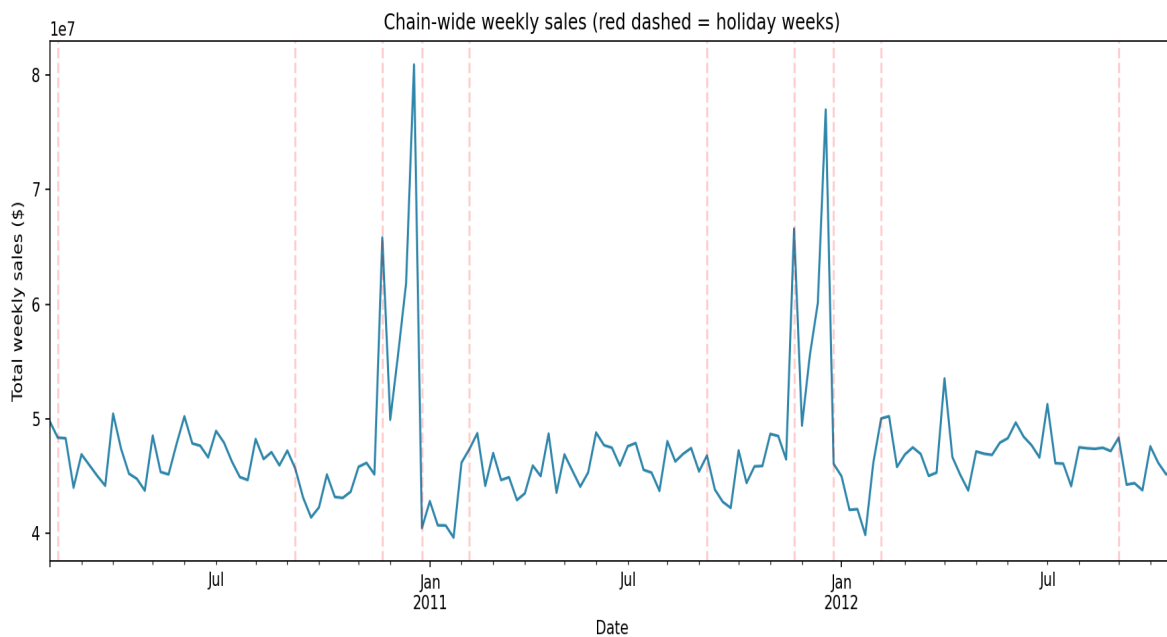


Fig 1. Chain-wide weekly sales — spikes are almost entirely Thanksgiving/Christmas weeks.

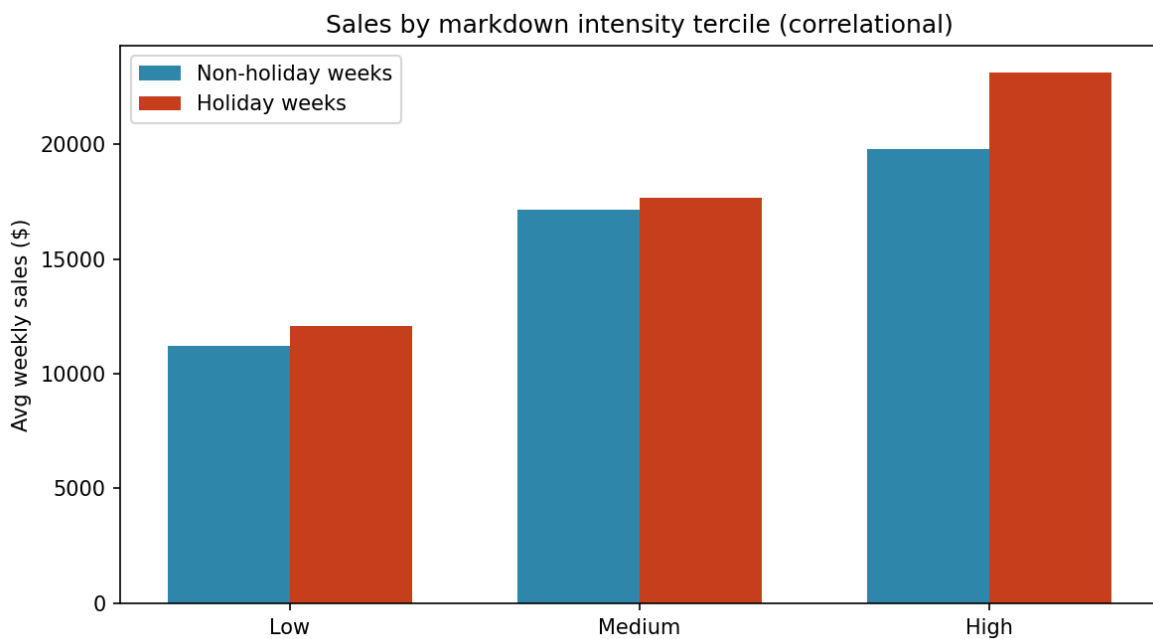


Fig 2. Sales by markdown intensity tercile, split by holiday flag — correlational, see So What above.

Method

- Merged train.csv (weekly sales) + features.csv (macro/markdown) + stores.csv (type, size) on Store/Date.
- Markdown data only exists from 2011-11-11 onward; comparisons involving markdowns restricted to that window.
- Promotion effect bucketed into Low/Medium/High terciles within holiday and non-holiday groups separately, to avoid holiday effect leaking into the promo read.

- Forecast: 12-week time-based holdout (no shuffling), WMAE with holiday weeks weighted 5x, matching the Kaggle competition's own scoring.

Recommendation

- Don't scale markdown spend chain-wide based on the raw +76-91% correlation — run a controlled test (matched stores/weeks) before committing incremental promo budget.
- Type C stores' sales-per-sqft is competitive with A/B once size is accounted for — expansion capital is better spent on more Type A-sized locations than assuming Type A's format outperforms.
- Use the moving-average baseline as the forecast floor for planning — a real model needs department-level seasonality since the 81 departments likely differ sharply in holiday sensitivity.

Source: Walmart Recruiting - Store Sales Forecasting (Kaggle). Full pipeline reproducible via [analysis.py](#); interactive version in [dashboard.html](#).